

As an aspiring data scientist I am acquiring data analysis and machine learning tools to take up ongoing astronomical challenges, and delve into the Universe. I am looking for opportunities to further grow my astrophysics knowledge, and test novel deep learning paradigms.

EDUCATION

MSc – Physics of Data

2024–Now

Department of Physics and Astronomy, University of Padua (Italy)

- **Current Grade Point Average** – 29.2/30

BSc – Astronomy

2019–2024

Department of Physics and Astronomy, University of Padua (Italy)

- **Final Grade** – 96/110
- **Thesis** – A/m distributions from in-orbit spacecraft fragmentation.
- **Supervisor** – Prof. Giampaolo Piotto — **Co-supervisor** – PhD Lorenzo Olivieri
- **Summary** – Study on the main factors determining A/m distributions of spacecraft fragments, and the importance of this parameter on the lifetime of these dangerous fragments in Earth orbits.

MSC STUDY PLAN

Data Analysis

- **Computer Vision** – Mark: 30/30
- **Machine Learning** – Mark: 28/30
- **Neural Networks and Deep Learning** – Mark: 30/30
- **Natural Language Processing** – Undertaking
- **Laboratory of Computational Physics** – ML algorithms implementation, applied to physics scenarios. Mark: 30/30
- **Management and Analysis of Physics Datasets** – CPU architecture, and distributed computing. Mark: 29/30
- **Modern Computing for Physics** – Algorithm acceleration using CUDA, and OpenMP. Undertaking
- **Advanced Statistics for Physics Analysis** – Mark: 28/30
- **Information Theory and Inference** – Undertaking

Physics

- **General Relativity** – Mark: 29/30
- **Compact Object Astrophysics** – Undertaking
- **Gravitational Physics** – Undertaking
- **Astronomical Interferometry** – Undertaking

SKILLS

Proficient

- **CLI** – Bash, Git, Conda
- **Python** – Jupyter, Dask, PyTorch, Keras, scikit-learn, Optuna, OpenCV, Pandas
- **R** – Tidyverse
- **MySQL**

Undertaking

- **C++** – CUDA, OpenACC, OpenMP
- **SNANA** – Supernova Analysis package

Languages

- **Italian** – Mother tongue
- **English** – B2

RELEVANT PROJECTS

Galaxy Classification using CNNs

GROUP PROJECT

- Implementation and comparison of several CNN architectures using PyTorch and Optuna to classify galaxy morphologies from the GalaxyZoo 2 dataset.
- The best architectures were able to consistently classify galaxies except for finer details such as weak lenses or bars.

Autoencoder for Gravitational Wave Detection

GROUP PROJECT

- Noise reconstruction, and anomaly detection via Long Short-Term Memory Autoencoder models, applied to a gravitational waves dataset.
- Several variations of the original model were tested, though none improved the reconstructions, they may represent an interesting baseline for further work.

Speed of Light Fast Fourier Transform - WIP

SOLO PROJECT

- Radix-2 Cooley-Tukey DFT implementation and benchmarking in serial, multicore and GPU cases, via OpenMP and CUDA.
- Further gaussian blurring in the frequency domain, and inverse FFT is performed.

Distributed processing of CORD-19 Analysis

GROUP PROJECT

- Accelerate NLP algorithms over a large dataset using Dask on a cluster of remote VMs on CloudVeneto servers.
- Focus on benchmarking multiprocessing vs. multithreading usage case in Python/Dask.

Segmentation and Classification of Skin Lesion Images

SOLO PROJECT

- Implement computer vision techniques via the OpenCV Python library to isolate lesions from the ISIC 2019 challenge dataset.
- Extract meaningful features to be fed to a Random Forest to classify the images.
- The classifier was able to identify the images with adequate accuracy for both considered classes.

SMC-ABC Sampling for Supernovae Ia Datasets - WIP

GROUP PROJECT

- Sequential Monte Carlo Approximate Bayesian Computation algorithm to build a posterior probability distribution for a set of cosmological parameters, using Supernovae Ia simulations obtained by the SNANA software package.
- Neural Likelihood-Ratio Estimation is also implemented as a comparison to the SMC-ABC algorithm.

Hierarchical Mergers of Binary Black Holes

GROUP PROJECT

- Random Forest application to uncover the main agents of longer merger chains using scikit-learn.
- The difference between the escape velocity and the gravitational kick resulted to be the main factor. The algorithm was not able to predict the final generation from the given data.

Multinomial Naive Bayes Classifier for Fake News Recognition

SOLO PROJECT

- MNB classifier implementation in R using a multilabel and a binary dataset to identify fake news.
- The algorithm struggled with similar samples as the ones presented in the multilabel dataset, while performed reliably on the binary one.

OBSERVATIONS AND RELATED EXPERIENCES

Physics of Data Workshop

24 SEPTEMBER 2025 - 26 SEPTEMBER 2025

Asiago Astrophysical Observatory (Italy)

- Attendance to several seminars about open research challenges in various physics fields through the lens of data modeling.

High Energy Astrophysics International Data Camp

25 SEPTEMBER 2023 - 29 SEPTEMBER 2023

University of Padua (Italy)

- Introduction and hands-on data analysis in high energy astrophysics using MAGIC data.

Observation Experience

24 NOVEMBER 2021

Asiago Astrophysical Observatory (Italy)

- Observation of NGC658 galaxy spectrum, and mass inference, for the BSc course “Astronomy Laboratory” final project.